



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : BIO F417
Course Title : Biomolecular Modeling
Instructor-in-Charge : SHIBASISH CHOWDHURY
Instructor : Jigneshkumar D Prajapati

1. Course Description: Biomolecular Conformations, Structural genomics and proteomics, protein folding, Forcefield, Simulation, Conformational analysis, ab initio structure prediction, comparative modeling, lattice models, usage of modeling packages.

2. Scope and Objective of the Course: The course is designed to provide students the firsthand experience of potential utility of biomolecular modeling especially in concurrent pharmaceutical research, and in cell and structural biology. It describes the functionality, advantages, and limitations of standard computing strategies for the simulation of biomolecules. Provide a working knowledge of freely available software to carry out independent research projects in biomolecular modeling and explore the possibilities for modeling to assist in the process of determination, analyzing, evaluating, displaying, and retrieving of 3D structure data in a research or industry laboratory environment.

3. Text Books: “Molecular modeling: Principles and Applications” By Andrew R Leach, 2nd Edition, 2001, Pearson Education Lim.

4. Reference Books: (1) “Molecular Modeling and Simulation - An Interdisciplinary Guide” By Tamar Schlick, Springer, New York, 2002

(2) “Principles of protein structure” By Schulz, G.E. and Schirmer, R.H., New York, Springer-Verlag, 1979

(3) “An Introduction to Computational Biochemistry” – By C. Stan Tsai, Wiley-Liss, Inc, 2002.

(4) “Bioinformatics: genes, proteins and computers” – Edited by C. Orengo, D. Jones, J. Thronton, BIOS Scientific Publishers Ltd., UK, 2003.

5. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
1 (1-2 Lecture)	Introduction	R-1 (1-2)	What is modeling? Scope and application of modeling in modern biology
2. (3-8 Lectures)	Protein structure and conformation	R-1 (3-4)	Amino Acid Building Blocks, Rotameric Structures Protein Conformation Framework, Ramachandran Plots, Conformational



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			Hierarchy, Structural motifs
3. (9-13 Lectures)	Conformation and Variability in DNA and RNA structures	R-1 (5-7)	Basic Building Blocks, Conformational Flexibility, Canonical DNA Forms, DNA Sequence Effects, Non-coding RNA structure and function, Challenges in RNA structure modeling and prediction
4. (14-17 Lectures)	Molecular graphics and Visualization	R-3 (4), Web resources	Introduction to graphic representation, Representation of molecular structure, Database of macromolecular structures, Usages of freely available visualization packages like VMD, ChimeraX, Rasmol, Pymol, SpdbViewer etc.
5. (18-24 Lectures)	Protein structure prediction and protein folding problem	R-4(8-9)	First principle methods for predicting protein structure, comparative modeling, threading, CASP, Protein folding theories, AlphaFold
6. (25-28 Lectures)	Energetics and Forcefield	T (4)	Different types of interactions and formulation of forcefield
7. (29-31 Lectures)	Molecular mechanics	T (5)	Basic algorithm of MM and their utilities, Hand on sessions
8. (32-35 Lectures)	Molecular dynamics	T (6-7)	Basic MD algorithm, Its limitation, treatment of long range forces, Enhanced sampling techniques
9. (36-38 Lectures)	Conformational analysis	T (9)	Analysis of molecular dynamics trajectories
10. (39-41 Lectures)	Lab on MD	GROMACS Online tutorials	Hand on session on molecular Dynamics

6. Evaluation Scheme:



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Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Comprehensive Examination	3 h	40%	07/12 FN	Partial Close Book
Assignment/seminar (several)		25%		Open Book
Surprise Quiz (several)		10%		Close Book/Open Book
Project		25%		Open Book

7. Grading Policy:

Award of grades would be guided by the histogram of marks and course average. **If a student obtains below 30% of the average of the top three performers or 40% of median, whichever is lower, the student will not be awarded a valid grade and the case will be reported as “NC”.**

8. Chamber Consultation Hour: To be announced in the class.

9. Notices: Notices, if any concerning the course will be displayed on the notice Board of Dept. of Biological Sciences.

10. Make-up Policy: Make up will be given on genuine grounds as determined by the Instructor-in-charge

11. Note (if any):

SHIBASISH CHOWDHURY
Instructor-in-charge
Course No. BIO F417